Vorhersage von ENSO-Ereignissen mit Hilfe eines STAR Modells

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Prediction of ENSO events using a STAR model

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Abstract

A linear auto regressive model, as well as two smooth transition auto regressive models are estimated for the Southern Oscillation Index, which is a commonly used index to measure El Niño - Southern Oscillation events. In order to compare the performance of the various estimated models, the data is divided into a training and a test data set. Both the logistic and the exponential auto regressive model with smooth transition show a lower accuracy in the test data set than the linear auto regressive model.

Contents

1 Introduction

1.1 Background

El Niño - Southern Oscillation, in short ENSO, is a climate phenomenon driven by interactions between the ocean and the atmosphere [Kli]. In the 19th century Peruvian fishermen gave the warming of the coastal waters towards the end of the year and the associated decline in fish stocks the name El Niño.

The second part goes back to the British meteorologist Sir Gilbert Walker, who described the interaction between the Southeast Asian low pressure area and the Southwest American high pressure area at the beginning of the 20th century. It was not until the middle of the 20th century that scientists recognized the connection between the two events. In the meantime, El Niño no longer stands for the warming of the Peruvian coastal waters alone, but for the entire eastern and central tropical Pacific.

As a fluctuation between unusually warm (El Niño) and cold (La Niña) conditions in the tropical Pacific, El Niño - Southern Oscillation is the most prominent year-to-year climate variation on Earth [Bal]. ENSO's fingerprints are imprinted on agriculture, fisheries, water resources, and economies across continents. Its influence affects atmospheric circulation patterns and triggers shifts in weather systems that are felt across the hemispheres.

In the context of ENSO, three states or phases of this tightly coupled ocean-atmosphere system are often distinguished in the Pacific region. The moderate and more frequent states are often described as the neutral phase, and the two extreme phases as El Niño and La Niña. It is not possible to say where exactly the transition of the phases is, it is more or less a steady process. One may speak of a neutral phase when there is an uneven distribution of air pressure over the Pacific. There is an area of high pressure over South America, which is stabilized by the low sea surface temperature (SST). The low SST is mainly caused due to the cold deep water that rises to the surface through the Humboldt Current and is well mixed due to the very high thermocline. At the same time, the air pressure over Australia and Indonesia is very low and trade winds try to compensate for this pressure difference. Among other things, the Humboldt and the South Equatorial Current are characterized by these winds. They take the surface water on an approximately 10.000 kilometer long journey from the coast of South America to Southeast Asia, meanwhile, the surface water warms up and therefore has on average 9 degrees more than in the West. Therefore, the area around the North of Australia and Indonesia is one of the largest rainfall areas in the world.

During an El Niño event, which occurs on average every 3 to 8 years, there is an increase in pressure over Southeast Asia and the Western Pacific for reasons that are still unclear, while the pressure in the Eastern Pacific off the coast of South America drops. As a direct consequence of the reduced pressure differential, the trade winds, which in principle transport the surface water of the Humboldt Current to the west, completely flatten out. In which case, the air pressure over Southeast Asia can be even higher than over the central Pacific, which is known as Southern Oscillation. At the same time, a weakening of the trade winds also has an impact on the sea surface temperature. During El Niño, both the sea level and the SST in the eastern Pacific rise, causing devastating floods in Southwest America, mostly Peru. Plankton and corals die due to the high water temperature, thus the food basis for fish and other marine animals fails and eventually also the fishery. On the same time the sea level and the SST drops in the Western Pacific causing heat periods that are unusual for this regions which lead to bushfires as well as droughts and dust storms. But much more, El Niño influences over three quarters of the climate on the whole earth. Many correlations are well demonstrable, while some remain only conjectures.

Despite the fact that La Niña is known as a counterpart for El Niño, the effects are not exactly the opposite. Characterising for a La Niña event is an even colder SST in the Eastern Pacific due to the increased upwelling of cold nutrient-rich deep water. In contrast to El Niño, the SST over the tropical west pacific is higher than normal. At the same time the air pressure is higher than normal over the tropical East Pacific and lower than normal over Indonesia and North America. Due to the even bigger difference in air pressure, the trade winds blow stronger than usual and create a large cold water tongue that reaches far to the west and provides less to non precipitation over Southwest America. Typically, La Niña occurs every 3 to 5 years and then usually lasts between 9 and 12 months on average. The consequences of La Niña are comparable in severity to the effects of an El Niño event. Besides, a La Niña event does not necessarily have to follow an El Niño event, and vice versa.

1.2 Aim of the work

As stated above, ENSO events influence ecosystems, agriculture, fisheries, and economies worldwide. In order to save human lives and prevent natural disasters from striking so suddenly, a meaningful forecast is very important. The goal of this work is to gain a better understanding of the differences between linear auto regressive models and nonlinear smooth transition auto regressive (STAR) models and to examine in more detail how these models perform in predicting ENSO events.

Greater certainty in predicting these events would only benefit the world. Authorities could provide more accurate and timely warnings about impending extreme weather events. This would enable better disaster preparedness, evacuation plans, and resource allocation to minimize the impacts of floods, droughts, and other disasters.

Also farmers could adjust their planting and harvesting schedules based on more reliable forecasts. This would reduce the risk of crop losses due to untimely droughts or excessive rainfall. Better prediction accuracy would also aid in selecting appropriate crop varieties and irrigation strategies.

A better understanding of how ENSO events might evolve under changing climate conditions would also contribute to more accurate climate change projections. This would enable policymakers to formulate more effective adaptation strategies that take both long-term climate change trends and short-term ENSO variability into account.

In essence, a better understanding of the dynamic behaviour of ENSO events would amplify the positive impacts of accurate forecasting, resulting in better resource management, reduced vulnerabilities and increased resilience to the adverse effects of ENSO events in a changing climate.

1.3 Organization and structure of the work

In the following chapter the basic STAR model, the most popular transition functions as well as a testing procedure for building a STAR model will be introduced. The third chapter introduces a definition of the Southern Oscillation Index and in general the data will be described and explained. The aim of the fourth chapter is to build a STAR model if we can reject linearity for the underlying data. Furthermore, the performance of the estimated models will be compared.

2 The STAR model

Smooth transition auto regressive models, in short STAR models, were first introduced and developed by Kung-sik Chan and Howell Tong in 1986 [CT86]. Originally they were called smooth threshold auto regressive models with the same acronym. In contrast to a normal auto regressive model the STAR model offers the additional possibility of describing a smooth transition between two (or more) regimes. This is achieved through a so-called transition function $G(s_t, \gamma, c)$ which is a continuous function bounded between 0 and 1. It allows for a smooth transition between regimes with are associated with different values of the transition function $G(s_t, \gamma, c)$. The regime at the time t is defined through the value of the observable variable s_t and the related value of the transition function $G(s_t, \gamma, c)$. Thereby the transition variable s_t can be a lagged endogenous variable, that is, $s_t = y_{t-d}$ for some integer $d > 0$ but it can also be any exogenous measurable variable. We will focus on the case with just two regimes and where s_t is endogenous. One regime will be associated with the transition function being equal to zero and the another regime with the transition function being equal to one.

2.1 Transition functions

The choice of the transition function depends on the regime-switching behaviour of the specific case [DF02]. We will present the most convenient types of transition functions with two regimes. Lets assume an uni-variate time series, lets say y_t, with observations at $t = 1-p, 1-(p-1), \ldots, 0, 1, \ldots, T$, where $p > 0$ is known as the order of the model. The STAR model is then given by

$$\begin{aligned} y_t = {} & \phi_{1,0} + \phi_{1,1}y_{t-1} + \phi_{1,2}y_{t-2} + \ldots + \phi_{1,p}y_{t-p} \\ & + (\phi_{2,0} + \phi_{2,1}y_{t-1} + \phi_{2,2}y_{t-2} + \ldots + \phi_{2,p}y_{t-p})G(s_t, \gamma, c) + \epsilon_t, \quad t = 1, \ldots, T \end{aligned} \tag{1}$$

or in a shorter version by

$$y_t = \phi_1' \tilde{y}_t + \phi_2' \tilde{y}_t G(s_t, \gamma, c) + \epsilon_t, \tag{2}$$

where $\phi_i' = (\phi_{i,0}, \phi_{i,1}, \ldots, \phi_{i,p}), i \in \{1, 2\}$ and $\tilde{y}_t = (1, y_{t-1}, y_{t-2}, \ldots, y_{t-p})'$. Thereby the error term ϵ_t is an independently identically distributed centered Gaussian process with finite second moment.

The first-order logistic function which is defined as

$$G(s_t, \gamma, c) = \big(1 + \exp(-\gamma(s_t - c))\big)^{-1}, \quad \gamma > 0 \tag{3}$$

is a popular choice for a transition function and the related model is named logistic STAR (LSTAR) model. The parameter c can be interpreted as a sort of threshold between the two regimes since the logistic function increases monotonically from zero to one in the observable variable s_t and equals 0.5 at $s_t = c$. The parameter γ can be thought of a smoothness parameter. For $\gamma \to \infty$ the transition function converges to the indicator function $h(s_t)$ with $h(s_t) = 0$ if $s_t < c$ and $h(s_t) = 1$ if $s_t > c$. Therefore the transition function changes immediately at $s_t = c$ and is far away from a smooth transition. For $\gamma \to 0$ the transition function approaches 0.5 in which case the LSTAR model reduces to a normal AR model. To sum up, in the LSTAR model the two regimes are associated with small and large values of s_t in comparison to c.

Sometimes it is more meaningful to connect the regimes to small and large absolute differences between the observable variable s_t and the parameter c. Therefore one might use the exponential function

$$G(s_t, \gamma, c) = 1 - \exp\big(-\gamma(s_t - c)^2\big), \quad \gamma > 0 \tag{4}$$

where the associated model is called exponential STAR (ESTAR) model. In this case the transition function approaches 1 both as $s_t \to \infty$ and $s_t \to -\infty$ whereas $G(s_t, \gamma, c) \to 0$ as s_t moves towards c. Nevertheless the ESTAR model collapses to a linear model when $\gamma \to \infty$ or $\gamma \to 0$.

A third type of transition function is the the second-order logistic function

$$G(s_t, \gamma, c) = \big(1 + \exp(-\gamma(s_t - c_1)(s_t - c_2))\big)^{-1}, \quad \gamma > 0,\ c_1 \leq c_2, \tag{5}$$

where $c = (c_1, c_2)'$. The model becomes linear for $\gamma \to 0$ same as the ESTAR model but for $\gamma \to \infty$ and $c_1 \neq c_2$ the transition function becomes the indicator function $h(s_t)$ with $h(s_t) = 0$ if $c_1 < s_t < c_2$ and $h(s_t) = 1$ otherwise, similar to the LSTAR model. In the case where $s_t = c_1$ or $s_t = c_2$ the transition function takes the value 0.5 but this should not bother us since it is just a λ-zero quantity. With this choice as transition function there are now two transitions points, at c_1 and c_2 respectively.

Based on the first and second order logistic function one can define the nth-order logistic function as

$$G(s_t,\gamma,c)=\Big(1+\exp\big(-\gamma\prod_{i=1}^{n}(s_t-c_i)\big)\Big)^{-1} \tag{6}$$

which can be utilized to achieve numerous transitions between the two regimes.

2.2 Modeling procedure

Before even applying a nonlinear model, like the STAR model, one has to be sure that the data of the underlying process is in fact nonlinear [Ter94]. The first step will be to build the linear auto regressive model. However, most of the time the order p will be unknown but the later specification of the STAR model builds on the assumption of knowing the auto regressive structure. Therefore one should use an appropriate model selection criterion such as the Akiake Information Criterion (AIC). In addition, it is important that there is little to no auto correlation in the model residuals otherwise the following tests will be a total miss-specification.

The null model is the selected linear model. Nevertheless, the null hypothesis $H_0:\gamma=0$ cannot be tested directly due to the unidentified nuisance parameter ϕ_2 in (2) which is not present in the null model. This is because the STAR model would reduce to a linear auto regressive model when $\gamma=0$ but also when $\phi_2=0$. Standard methods for these hypothesis tests are no longer meaningful enough [Dav87]. As a solution an approximation of the transition function with a third order Taylor expansion was proposed in [LST88] which leads to the auxiliary regression, expressed in (7):

$$y_t=\phi_0'\tilde{y}_t+\sum_{j=1}^{p}\beta_{2j}'y_{t-j}s_t+\sum_{j=1}^{p}\beta_{3j}'y_{t-j}s_t^2+\sum_{j=1}^{p}\beta_{4j}'y_{t-j}s_t^3+\epsilon_t \tag{7}$$

An equivalent null hypothesis for linearity and therefore against the STAR model is now

$$H_0':\beta_{2j}=\beta_{3j}=\beta_{4j}=0,\quad j=1,\dots,p \tag{8}$$

which can be tested with traditional testing methods.

Since we chose the transition variable s_t as a lagged endogenous variable y_{t-d} one may wonder what value is associated with the parameter d. Mostly this parameter is *a priori* unknown, but it is inevitable for our STAR model. A similar problem was discussed in [Tsa86] during the model specification. The idea came up to run several linearity tests at the same time and use a different value for the parameter d each time. The value of d associated with the smallest of all p-values of the linearity tests should then be chosen as the delay parameter for the STAR model under the condition that at least one p-value is sufficiently small otherwise linearity would not be rejected.

If so, the next step is to choose a proper transition function. A sequence of three ordinary F-tests that can be used to get an indicator whether the transition function should have a logistic or exponential form was presented in [Ter94]. Let $\beta_i = (\beta_{i1}, \ldots, \beta_{ip})$ for $i \in \{2, 3, 4\}$. The first F-test ($F4$) should be

$$H_{01} : \beta_4 = 0 \quad \text{vs.} \quad H_{11} : \beta_4 \neq 0 \tag{9}$$

The second F-test ($F3$) which should be performed is

$$H_{02} : \beta_3 = 0 | \beta_4 = 0 \quad \text{vs.} \quad H_{12} : \beta_3 \neq 0 | \beta_4 = 0 \tag{10}$$

And the third F-test ($F2$) of the three nested ones is

$$H_{03} : \beta_2 = 0 | \beta_3 = \beta_4 = 0 \quad \text{vs.} \quad H_{13} : \beta_2 \neq 0 | \beta_3 = \beta_4 = 0 \tag{11}$$

In general a rejection of 9 or 11 can be interpreted as a rejection of the ESTAR model, while not rejecting 10 is in favor of choosing the LSTAR model. As a rule of thumb, [Ter94] recommended to choose an ESTAR model, when the p-value of F3 is smallest and an LSTAR model otherwise.

2.3 Estimation of parameters

After determining the structure of the STAR model one has to estimate the parameters. This can be done with the nonlinear least squares method. Sometimes the algorithm does not converge, this is mostly because of to much complexity in the model. To overcome this problem one might use a grid for especially the non-linear parameters and minimize the residual sum of squares while these parameters are kept fixed. If this does not help at all, parameter restrictions could be taken into account in order to reduce the number of parameters to be estimated. For example, $\theta_{10} = -\theta_{20}$ would be a reasonable constraint, in which case the intercept would not contribute anything to the process in the 'normal' regime when $G(s_t, \gamma, c) = 1$.

If the STAR model has been specified, the validity of the model must be evaluated. First of all, one should check whether the estimates look reasonable, because of the existence of local minima, especially in short time series. Indicators for an unsatisfactory model might be when the parameter c is far outside the range of the observed variables y_t, or when the model diverges in the long run. This can be done numerically by simply selecting observed values as starting values and extrapolating the time series without adding white noise. A satisfactory model must converge. Also an inspection of the model residuals and their auto-correlations should be done. We refer to [Ter94] for some test examples.

3 Data description

A strong indicator in order to describe El Niño - Southern Oscillation events is the Southern Oscillation Index, in short SOI, which is calculated based on the difference in the air pressure between Tahiti and Darwin [Env]. Among many slightly different definitions of the SOI, we will use the one presented by the National Center for Environmental Information. Lets start with some definitions:

$$\sigma_1 = \sqrt{\sum_{i=1}^{N} \left(\mathrm{aSLP}_i^j - \mathrm{mSLP}^j\right)^2 / N}, \quad j \in \{\text{Tahiti, Darwin}\} \tag{12}$$

aSLP_i^j is the averaged actual sea level pressure for the month i in Darwin or Tahiti and mSLP^j is the mean sea level pressure of all aSLP's in the time period from 1981 to 2010. N is the number of observed months. Furthermore, one can define the standardized sea level pressure for a particular month i:

$$\mathrm{sSLP}_i^j = \frac{\mathrm{aSLP}_i^j - \mathrm{mSLP}^j}{\sigma_1} \tag{13}$$

For further standardization purposes define

$$\sigma_2 = \sqrt{\sum_{i=1}^{N} \left(\mathrm{sSLP}_i^{\text{Tahiti}} - \mathrm{sSLP}_i^{\text{Darwin}}\right)^2 / N}. \tag{14}$$

The SOI for a particular month i is then calculated as

$$\mathrm{SOI}_i = \frac{\mathrm{sSLP}_i^{\text{Tahiti}} - \mathrm{sSLP}_i^{\text{Darwin}}}{\sigma_2} \tag{15}$$

Monthly SOI data from January 1951 is available from the National Center for Environmental Information and will be used in this study. In general, the SOI time series agrees very well with changes in ocean temperatures, and thus describe the extreme events very well. A positive SOI indicates a La Niña event as atmospheric pressure is higher than usual over Tahiti and lower than usual over Darwin, resulting in stronger trade winds. Conversely, a negative SOI means lower pressure difference between east and west and thus weak to no trade winds which are typically for El Niño.

When looking at the SOI in figure 1 one might see a devastating El Niño event in 1982/83 and the most recent countable event in 2015/16 whereas strong La Niña's happened in 1973/74, 2007/08 and 2010/11.
Furthermore, the auto covariance function approaches zero. Therefore, there might be some time dependence in the data, but it decreases with the lag and eventually becomes negligible. A plot of the auto-covariance function can be seen in figure 2.

4 Empirical work

The aim of this chapter is to build a linear auto regressive model based on the given data with the help of the statistical program R and furthermore use it as the null model for testing against linearity [R C21]. If linearity is rejected, the next step is to build a STAR model as described in the second chapter. The selected models are then compared in terms of their accuracy.

4.1 Testing against linearity using the linear model

For the following steps we will only use data until December 2000 to estimate the parameters of the models. This is to ensure that we have enough data to verify and compare the performance of the models afterwards. We will speak of them as training and test data set. As a first step it is necessary to determine the linear model. For this purpose we will use the ar-function which is already implemented in R. We choose the AIC as a model selection criterion and the ordinary least squares method to determine the parameters. According to the resulting coefficients the estimated linear auto regressive model is given by

$$y_t = 0.0008 + 0.4639 y_{t-1} + 0.2313 y_{t-2} + 0.0740 y_{t-3} + \epsilon_t \tag{16}$$

A visual representation from the performance of the linear model can be seen in figure 3. Especially in the extreme periods of the process, more accuracy would be desirable. An analysis of variance between the linear model and the one from 7 was performed for each delay parameter $s_t = y_{t-d}$ with $1 \leq d \leq 3$ and afterwards concluded that we reject H_0' in 8 only for $d = 3$. Therefore, we reject linearity and choose y_{t-3} as our delay parameter s_t.

The next step in the earlier mentioned modeling procedure is to determine whether an LSTAR or ESTAR model should be used. For this purpose we performed three F tests specified in [Ter94] and based on the results which can be found in Table 1 an ESTAR model should be best for the data.

4.2 Estimation of parameters for the nonlinear model

To determine the coefficients of the ESTAR model, we used different strategies. First, we tried to take advantage of the nonlinear least squares method in R [R C21]. But due to the high complexity of the STAR model, the algorithm is not convergent. Therefore, we had to adapt the strategy in order to get a result. A grid was used to keep the parameters fixed which are essential for the non-linearity and calculate all the other parameters with the ordinary least squares method.

Therefore, we used a grid for γ and c and estimated the other parameters. The corresponding ESTAR(3) model is the following:

$$\begin{aligned} y_t &= 1.122 + 2.902y_{t-1} - 0.848y_{t-2} - 0.992y_{t-3} \\ &\quad + [-1.111 - 2.467y_{t-1} + 1.085y_{t-2} + 1.121y_{t-3}] \times \left[1 - \exp\left(-3.936(y_{t-3} + 3.086)^2\right)\right] \end{aligned} \tag{17}$$

$$n = 600,\ s = 0.672,\ s_{\text{nonlinear}}/s_{\text{linear}} = 0.984$$

In this case, n is the sample size, s is the estimated standard error of the residuals and s_{linear} the one from the linear model. The ratio between the standard deviations then gives an indication of how much better the ESTAR model describes the data in relative terms. Since the ratio is close to unity one might assume that non-linearity is only needed in the most turbulent periods.

A comparison between the measured values and the ones estimated by the ESTAR model can be seen in figure 4. Although there are still deviations from the observed values, the ESTAR model describes the extreme phases a little better than the linear model, especially El Niño events.

Figure 5 shows the shape of the transition function $G(y_{t-3}, \gamma, c)$ in the delay variable y_{t-3} - the larger the cross, the more observations are in the data with that specific value of the SOI. Figure 6 shows the same information but with the values of the delay variables ordered over time while the grey graph in the background represents the estimated values from the ESTAR model. Both figures show that the transition function is mostly close to unity and since the intercepts θ_{10} and θ_{20} sum up close to zero, the intercept term contributes to the process only when $G < 1$, which is quite a rare event. In fact, in more than 98% of the 600 monthly averages of the SOI, the transition function is equal to 0.95 or higher.

4.3 Validation of the ESTAR model

When looking at 17 and figure 1 it can be seen that the parameter $c = -3.086$ is on the lower side of the observed values for the SOI. Therefore the regime, where the transition function G is near zero, will be interpreted as an El Niño event.

When extrapolating the time series with several different sequences of start values, the series converges always, which can be interpreted as a sign for a satisfactory model. Nevertheless, when having a look at the performance between the models on the test data set which includes the monthly SOI from 2001 until 2023 the linear model has a lower residual sum of squares. A reason could be over-fitting of the ESTAR model on the training data set. Additionally, we plotted the auto correlation function for the test data set of both models, but no unusual behaviour has been detected. The result of the performance tests can be found in table 2. To sum it up, the ESTAR model in 17 does not fulfill our goal.

4.4 Alternative models

Since we didn't want to settle for that, we decided to build a LSTAR model as well. The performance on the training data set was slightly better than the one from the ESTAR model and therefore, much better than the linear model when speaking of the residual sum of squares and the estimated standard deviation. The calculated LSTAR(3) model was the following:

$$\begin{aligned} y_t = & -2.029 + 0.696 y_{t-1} + 0.351 y_{t-2} - 1.123 y_{t-3} \\ & + [2.057 - 0.269 y_{t-1} - 0.127 y_{t-2} + 1.238 y_{t-3}] \times [1 + \exp(-58.554(y_{t-3} + 1.413))]^{-1} \end{aligned} \tag{18}$$

$$n = 600,\ s = 0.669,\ s_{\text{nonlinear}}/s_{\text{linear}} = 0.981$$

We did not include a visual comparison between the fitted values of the LSTAR model and the observed values since it looked quite similar to the one from the ESTAR model. Corresponding figures for the transition function can be found in the figures 7 and 8. Similar tests as for the ESTAR model have been carried out for the validation of the LSTAR model on the training data set and nothing unusual has been noticed. But same as for the ESTAR model, the performance on the test data set was worse then the one from the linear model. The corresponding numbers can be found again in table 2.

Furthermore, we had a closer look on the residuals from the three estimated models on the test data set. A plot of the cumulative sum of the residuals can be found in figure 9. One can see that the ESTAR model describes the strong El Niño in 2005 worse than the linear model. The LSTAR model has problems as well at this specific event and also for the El Niño in 2016. Apart from these two cases the models perform nearly the same.

We also tried to create a STAR model for other delay parameters d (i.e. $d = 1$ and $d = 2$). Despite the strong acceptance of linearity for $d = 2$, the LSTAR model performed best in terms of residual sum of squares compared to all previously created models. Nevertheless, the residual sum of squares was much higher in the test data set, indicating even higher over-fitting.

5 Conclusion

It is clear that STAR models will always describe the data on which they were estimated better than the linear model. There is more flexibility, which represents a small advantage over the linear model, particularly in the more extreme phases of the time series. Still, one might argue if a reduction of the residual sum of squares by approximately 1.6 percent in the ESTAR(3) and 1.9 percent in the LSTAR(3) model can be interpreted as a significant improvement. Either way, both STAR models disappoint on our test data set in comparison to the linear model, whereas we will not use them to predict ENSO events.

Also, one can say for sure that describing ENSO activity is by far depending on more than just pressure differences between Tahiti and Darwin. Since in our case the STAR models gave the impression of over-fitting one might use an exogenous variable as the transition variable s_t instead of a lagged endogenous variable.

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List of Figures

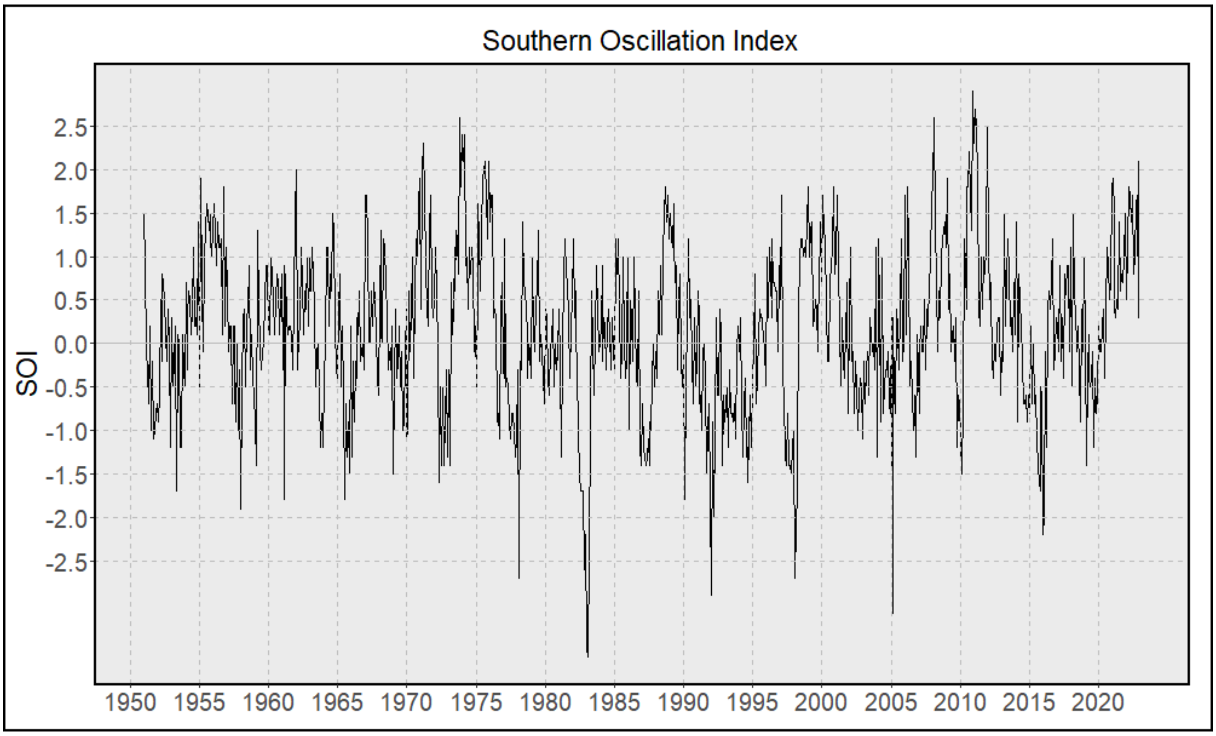

Figure 1: Time series of the Southern Oscillation Index

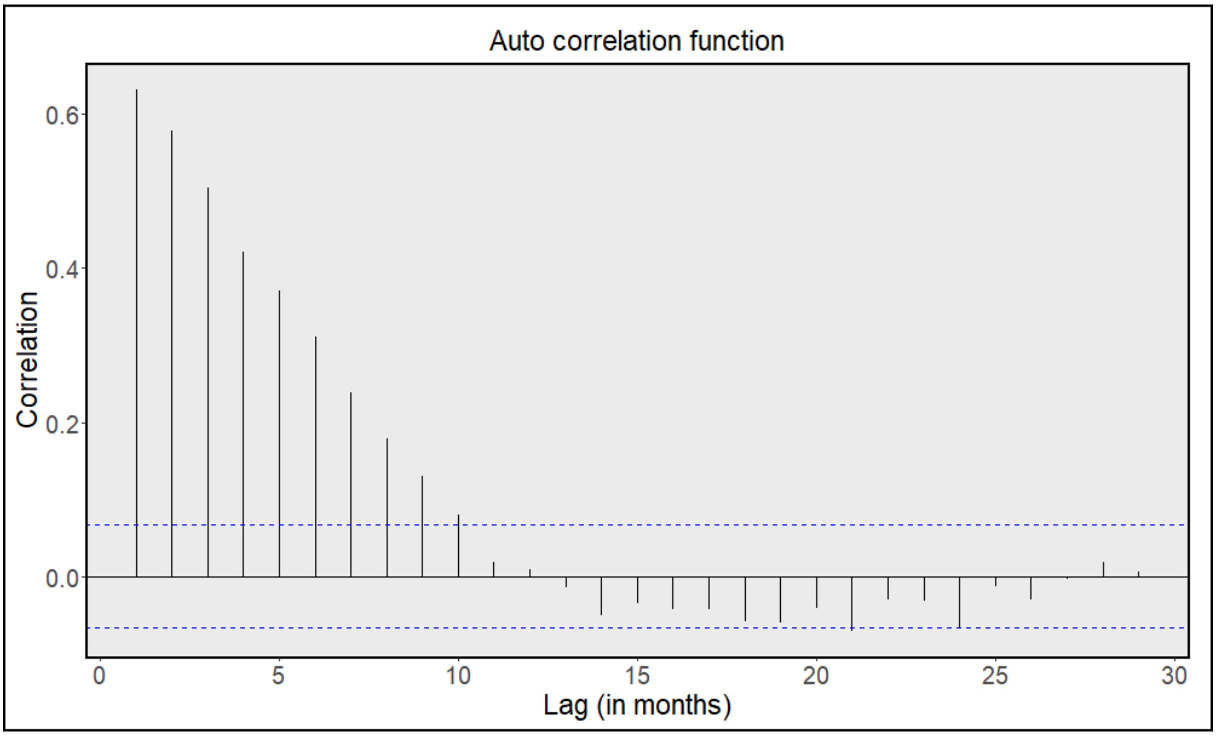

Figure 2: Auto correlation function of the Southern Oscillation Index

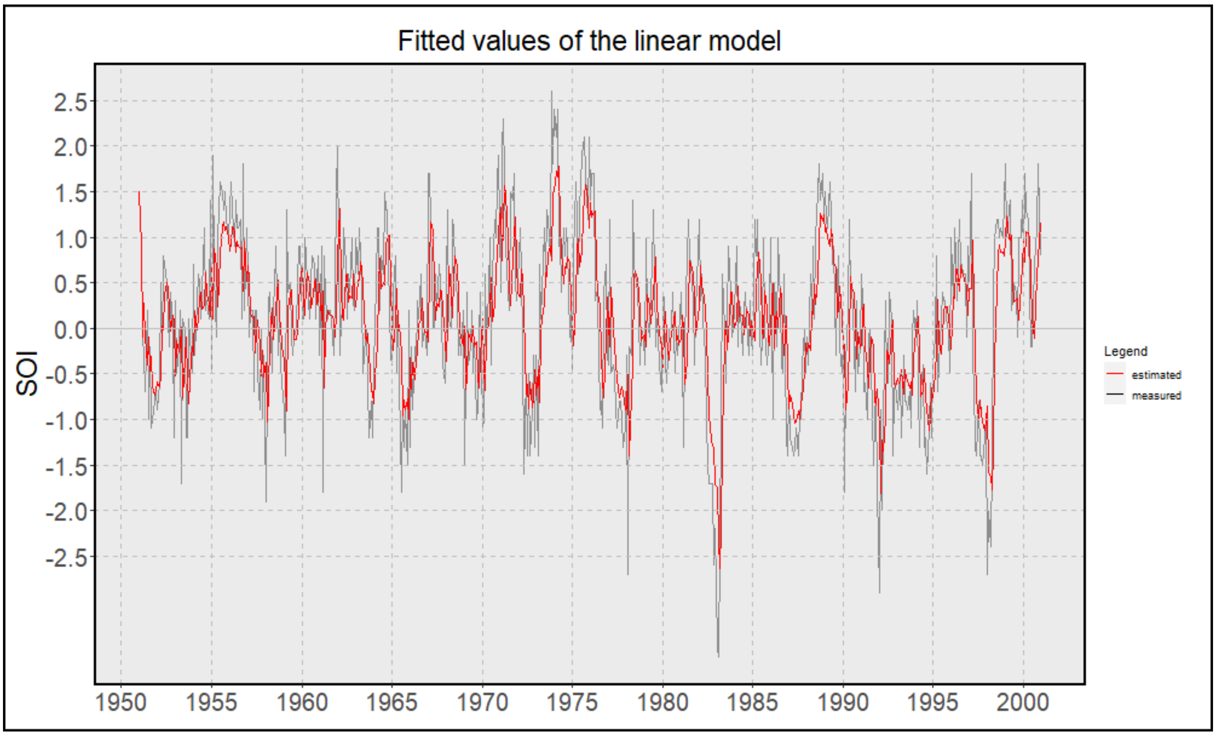

Figure 3: Observation vs. Estimation: SOI in an AR Model

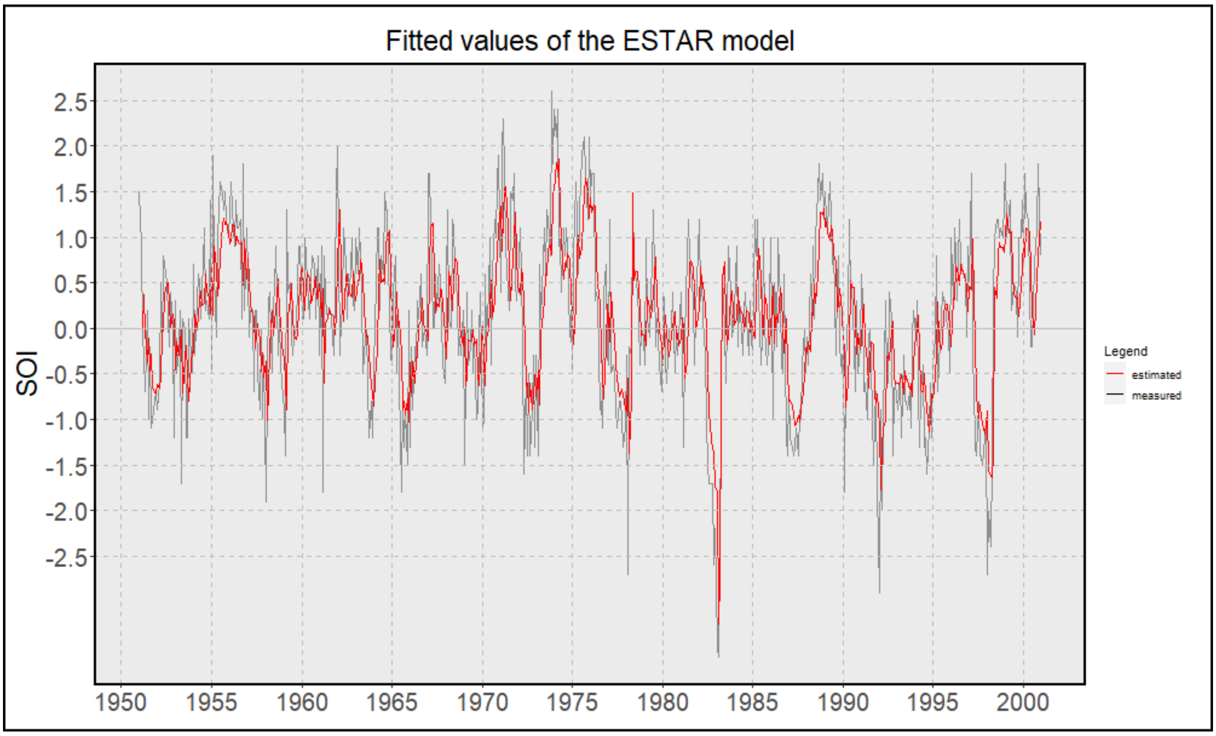

Figure 4: Observation vs. Estimation: SOI in an ESTAR model

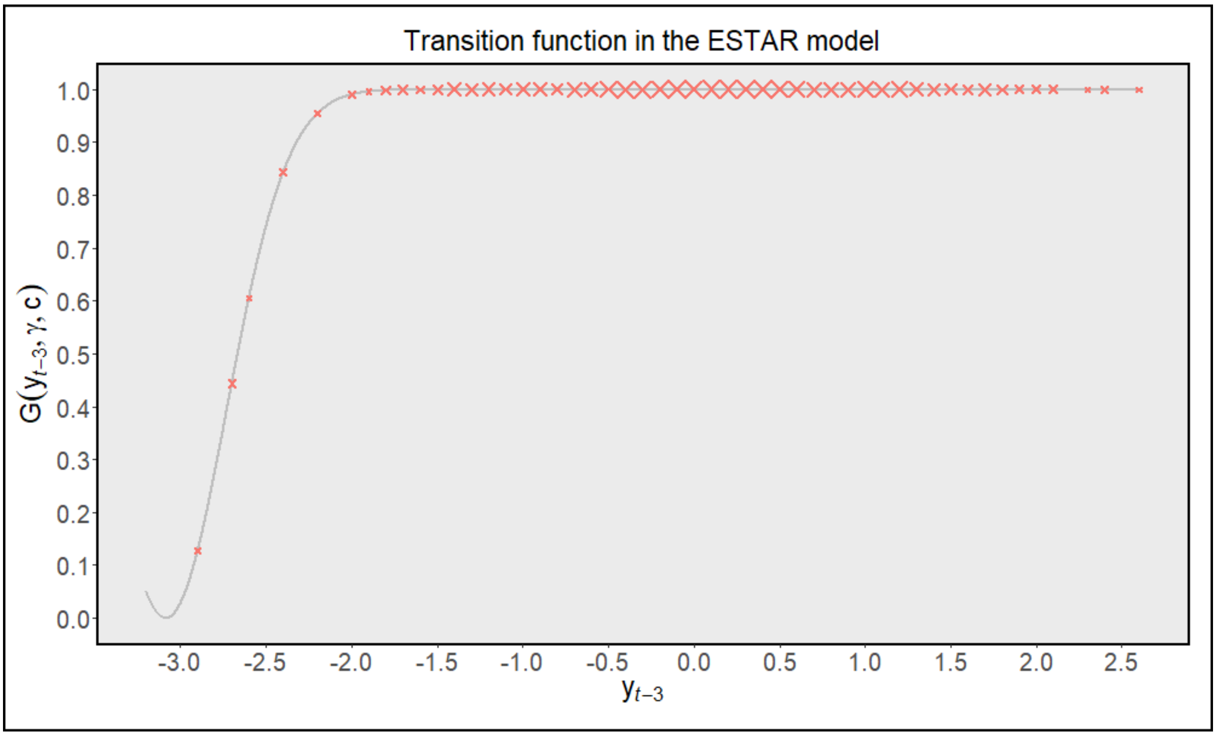

Figure 5: The graph of the transition function in the delay variable y_{t-3} (ESTAR)

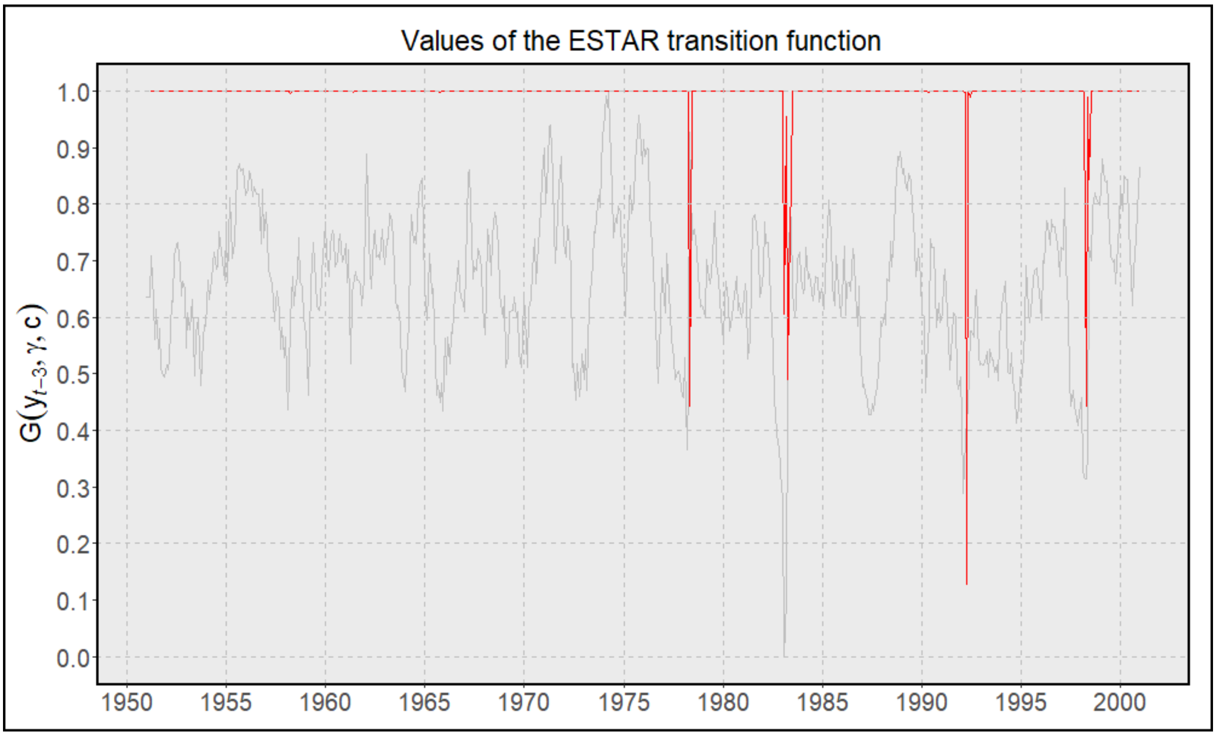

Figure 6: Values of the transition function in the ESTAR model during the observed period

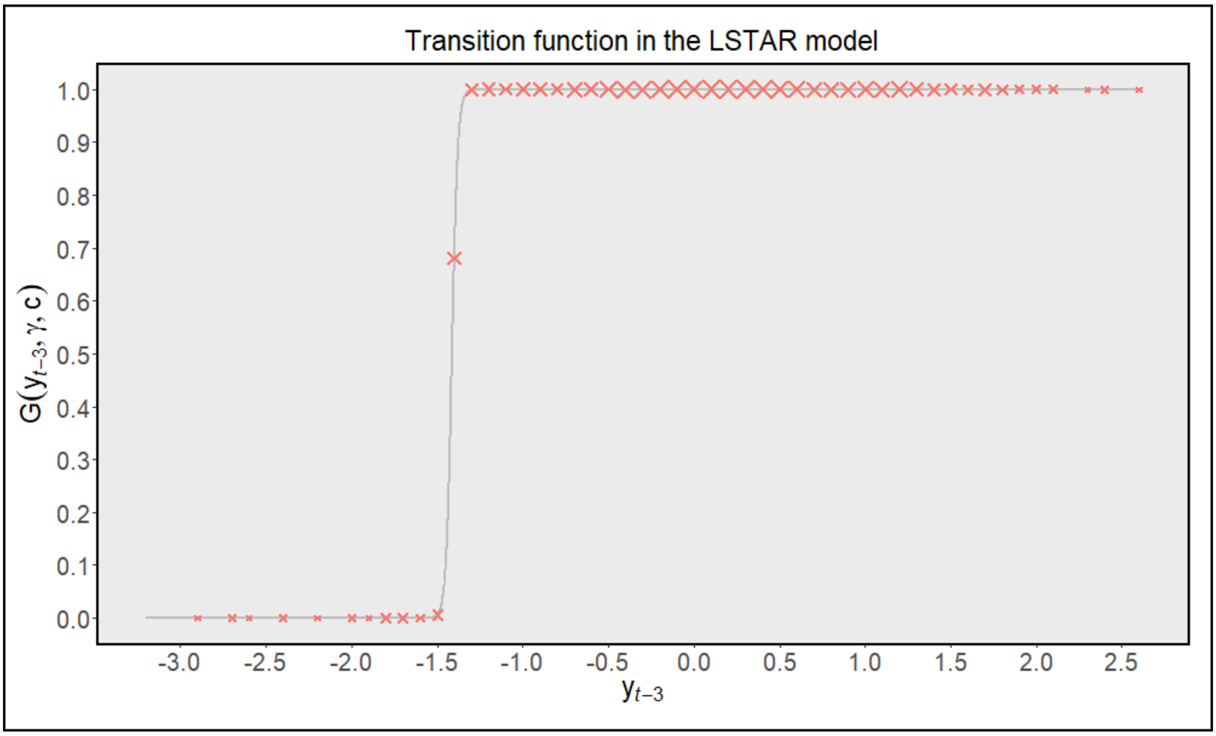

Figure 7: The graph of the transition function in the delay variable y_{t-3} (LSTAR)

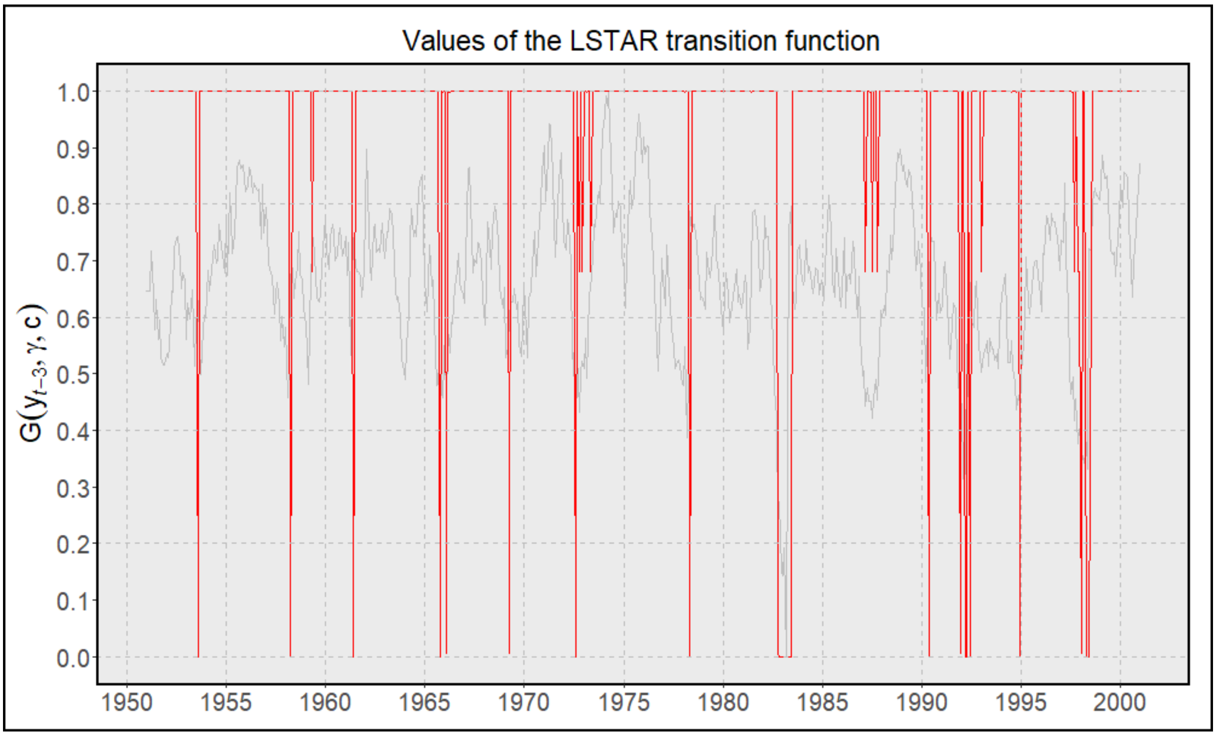

Figure 8: Values of the transition function in the LSTAR model during the observed period

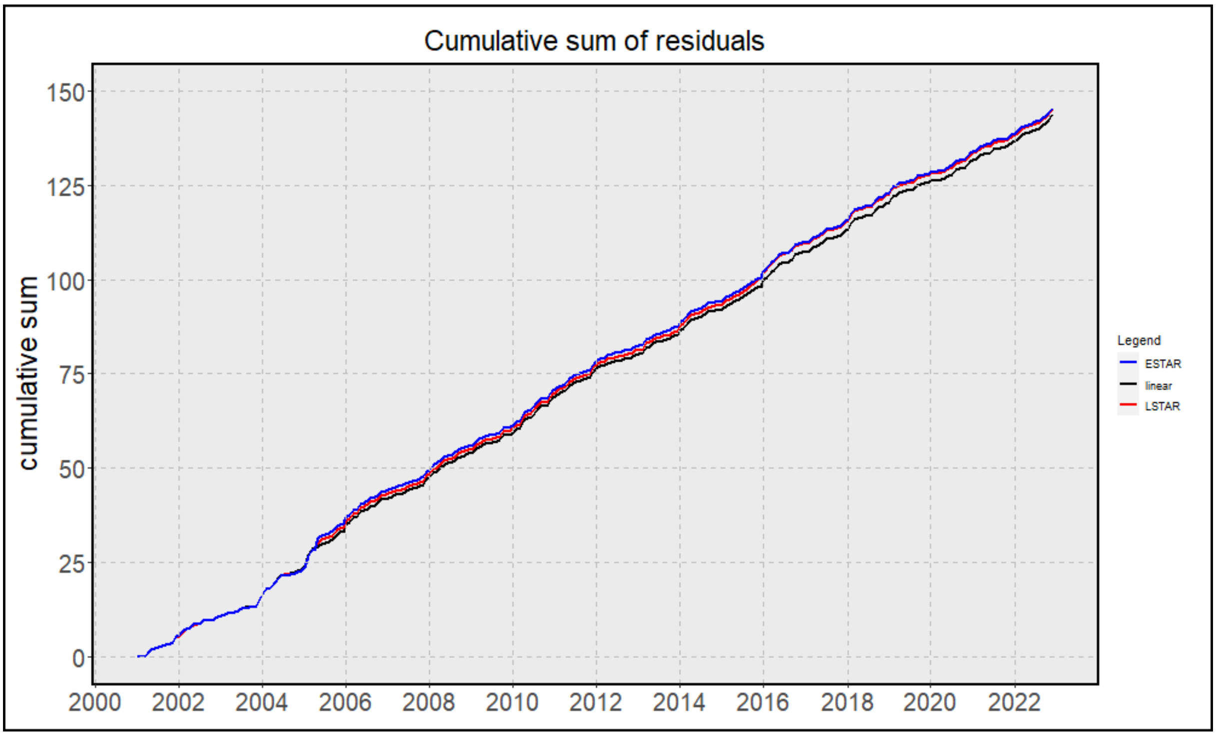

Figure 9: Cumulative sum of the residuals on the test data set for the three estimated models

p	d	$p(F_L)$	$p(F_4)$	$p(F_3)$	$p(F_2)$	model
3	1	0.9250				
3	2	0.3213				
3	3	0.0011	0.6258	$1.48 \cdot 10^{-4}$	0.1450	ESTAR

Table 1: Results of linearity tests and model selection

Here p is the order of the estimated linear model, d is the delay factor, $p(F_L)$ is the p-value of the linearity test, the following three columns are the p-values for the F tests for the model selection and the last column states whether an ESTAR or LSTAR model should be used. An ESTAR model is selected since $p(F_3)$ is by far the smallest of the three F tests.

model	training data set		test data set	
	RSS	sd	RSS	sd
linear	279.272	0.682	132.305	0.626
ESTAR	267.153	0.672	139.335	0.650
LSTAR	265.0614	0.669	134.243	0.622

Table 2: Residual sum of squares and standard deviations for the linear, ESTAR and LSTAR model calculated on the training and test data set